COMPACT HODGE TRACE STRATIFICATION IN THE CALABI–YAU EXAMPLES

RAEEZ LORGAT

Abstract. The compact Hodge supertrace

$$\Xi(X) = \sum_q (-1)^q h^{0,q}(X)$$

is the compact numerical invariant of the Calabi–Yau examples. Its interpretation as κ_{ch} is a theorem on the constructed Φ_d-loci and a construction problem beyond them. The table through dimension five separates this compact invariant from the noncompact local surface value, the conifold NCCR value, the Borcherds weight, and the fibre Euler characteristic. These numbers come from different constructions and must carry their subscripts.

1. Four invariants

Definition 1. For a compact Calabi–Yau X define

$$\Xi(X) = \sum_q (-1)^q h^{0,q}(X).$$

On the compact loci where the trace comparison for $A_X = \Phi_d(\mathrm{Perf}(X))$ has been constructed, this gives $\kappa_{\mathrm{ch}}(A_X) = \Xi(X)$. The $d \geq 4$ interpretation remains conditional. Heisenberg, BKM, fibre, BCOV, and noncompact toric specialisations are separate invariants. The categorical invariant is

$$\kappa_{\mathrm{cat}}(X) = \chi(\mathcal{O}_X),$$

with Künneth-multiplicative behaviour on products. The Borcherds invariant is

$$\kappa_{\mathrm{BKM}}(\Phi_N) = \frac{c_N(0)}{2},$$

on the CHL/Gritsenko–Nikulin Borcherds-product levels $N \in \{1, 2, 3, 4, 6\}$ with the chosen weak Jacobi input. The Cléry–Gritsenko diagonal-divisor catalogue is a different Borcherds-product catalogue. In its triple order the weights are

$$\left(5, 2, 3, 1, 2, \frac{1}{2}, \frac{3}{2}, 1\right),$$

with integer-weight paramodular rows and two half-integer multiplier-system rows. These weights are not the CHL sequence $(5, 4, 3, 2, 1)$. The fibre invariant κ_{fiber} is the topological Euler/Mukai-lattice rank of the chosen fibre; for a K3 fibre it is 24, not $\chi(\mathcal{O}_{K3}) = 2$.

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2. Compact dimension table

Theorem 2. *The Hodge supertrace gives the following unconditional compact table:*

d	X	$(h^{0,0},\dots,h^{0,d})$	$\Xi(X)$
1	E	(1, 1)	0
2	$K3$	(1, 0, 1)	2
2	*abelian surface*	(1, 2, 1)	0
3	*quintic threefold*	(1, 0, 0, 1)	0
3	$K3\times E$	(1, 1, 1, 1)	0
3	E^3	(1, 3, 3, 1)	0
4	*sextic fourfold*	(1, 0, 0, 0, 1)	2
5	*generic strict* CY_5	(1, 0, 0, 0, 0, 1)	0

Proof. For a strict compact Calabi–Yau d-fold, Serre duality and the standard Hodge-column vanishing leave only $h^{0,0}$ and $h^{0,d}$, giving $1+(-1)^d$. Product examples follow by Künneth on $h^{0,*}$; this gives the $K3\times E$ and E^3 rows. The abelian-surface row is the exterior algebra on $H^1(\mathcal{O})$. The hypersurface rows use Lefschetz and Kodaira vanishing. The displayed values are the resulting Hodge supertraces. The constructed-locus theorem identifies these numbers with κ_{ch} only where the object $A_X=\Phi_d(\mathrm{Perf}(X))$ and its trace comparison have been built; the $d\geq 4$ rows remain compact Hodge data until that construction is supplied. $\square$

3. Local and singular examples

Proposition 3 (Local $\mathbb{P}^2$). *For the local surface* $X=\mathrm{Tot}(K_{\mathbb{P}^2}\to\mathbb{P}^2)$ *the chiral shadow normalisation gives*

$$\kappa_{\mathrm{ch}}(X)=\frac{3}{2}.$$

Proof. The local $\mathbb{P}^2$ shadow theorem computes the same value by the $\mathbb{Z}/3$-McKay quiver datum for $\mathbb{C}^3/\mathbb{Z}_3$, by three affine toric charts each contributing $1/2$, by the MacMahon exponent $M(q)^{\chi_{\mathrm{top}}(\mathbb{P}^2)}$, and by the compactly supported Dolbeault trace on the base. The trace comparison carries the local compact-support normalisation, which contributes the factor $1/2$. Thus the local-surface value is $\chi_{\mathrm{top}}(\mathbb{P}^2)/2=3/2$. This is a noncompact CY_3 local-surface reading, separate from the compact Hodge supertrace table. $\square$

Proposition 4 (Conifold). *The resolved conifold*

$$\mathrm{Tot}(\mathcal{O}(-1)\oplus\mathcal{O}(-1)\to\mathbb{P}^1)$$

is outside the local surface family $\mathrm{Tot}(K_S\to S)$*. The conifold NCCR calculation gives*

$$\kappa_{\mathrm{ch}}(\mathrm{conifold})=1.$$

Proof. The resolved conifold is the small resolution over $\mathbb{P}^1$, outside the canonical-bundle-over-a-surface family. The finite-quotient BKR McKay route for $\mathbb{C}^3/G$ is inapplicable: the conifold is the affine hypersurface $xy-zw=0$, and its crepant resolution is governed by the Szendroi/Klebanov–Witten noncommutative crepant resolution, equivalently the two-vertex conifold quiver with potential. The value 1 is the direct conifold NCCR/Hochschild shadow calculation: the level-one DT/CoHA coefficient on the compact class $[\mathbb{P}^1]$ is 1. The half-Euler value of the exceptional curve is only a numerical coincidence here. $\square$

Proposition 5 (Affine positive half and toric gluing)**.** *For the affine chart $\mathbb{C}^3$, with the three-loop Jordan quiver and potential $W = \operatorname{tr}(XYZ - XZY)$,*

$$\mathrm{CoHA}^{\mathrm{sph}}_{\mathrm{crit},T}(\mathbb{C}^3, W) \cong Y^+(\widehat{\mathfrak{gl}}_1).$$

The target is the positive half. It is not $\mathcal{W}_{1+\infty}$. A finite toric CY_3 atlas with simplex-compatible DWR/Ran Rees hCS–Hall data glues these local positive halves to

$$H^+_{X_\Sigma} = \mathrm{hocolim}_{\alpha\in\Sigma(3)} \mathrm{CoHA}^{\mathrm{sph}}_{\mathrm{crit},T}(U_\alpha).$$

Compact critical CoHA, quasi-NCCR, and Hall–Drinfeld double outputs are separate compact realisation problems: they require monoidal vanishing-cycle realisation, compact support, a non-degenerate pairing after radical quotient, completion, opposite and Cartan halves, and PBW/no-extra-relations compatibility.

Proof. Schiffmann–Vasserot identify the spherical equivariant critical Hall algebra of the tripled Jordan quiver with the shuffle presentation of $Y^+(\widehat{\mathfrak{gl}}_1)$. The DWR/Ran Rees comparison is formed chartwise from finite cyclic critical atlases; Stokes' formula on the simplex variables gives the face compatibility, and Hall wall-crossing quasi-isomorphisms identify the overlaps. Taking the finite homotopy colimit therefore gives the displayed positive half. None of these operations constructs the continuous dual, Hopf pairing, compact-support vanishing-cycle functor, or completed double needed for the compact objects. □

4. Borcherds calculation

Theorem 6 (Borcherds weight)**.** *For the CHL/Gritsenko–Nikulin Borcherds-product levels $N \in \{1, 2, 3, 4, 6\}$ with weak Jacobi input ϕ_N,*

$$\kappa_{\mathrm{BKM}}(\Phi_N) = \frac{c_N(0)}{2}.$$

The constants are

N	1	2	3	4	6
$c_N(0)$	10	8	6	4	2
$\kappa_{\mathrm{BKM}}(\Phi_N)$	5	4	3	2	1

Proof. In Borcherds' singular theta-lift normalisation, the automorphic product attached to a weight-zero weak Jacobi input has weight equal to one half of the constant coefficient of that input. In the CHL/Gritsenko–Nikulin levels this coefficient is $c_N(0)$, giving the displayed table. At $N = 1$ the input $\phi_{0,1}$ has $c_1(0) = 10$, so the theta-normalised Igusa square root has weight 5. The denominator identity reads that weight as κ_{BKM}. This calculation belongs to the automorphic Borcherds construction and is independent of the compact Hodge supertrace. Cléry–Gritsenko diagonal-divisor rows use a separate catalogue; the comparison tuple $(5, 2, 1, 1, \frac{1}{2}, 1, \frac{1}{4}, 0)$ is not that catalogue, and no Cléry–Gritsenko row has weight $\frac{1}{4}$ or 0. □

Proposition 7 (Cléry–Gritsenko diagonal-divisor rows). *The Cléry–Gritsenko classification gives exactly eight diagonal-divisor genus-two Borcherds products:*

(t, N)	$F_{t,N}$	$\mathrm{wt}(F_{t,N})$	$2\,\mathrm{wt}(F_{t,N})$
$(1,1)$	Δ_5	5	10
$(2,1)$	Δ_2	2	4
$(1,2)$	∇_3	3	6
$(3,1)$	Δ_1	1	2
$(1,3)$	∇_2	2	4
$(4,1)$	$\Delta_{1/2}$	$\frac{1}{2}$	1
$(1,4)$	$\nabla_{3/2}$	$\frac{3}{2}$	3
$(2,2)$	Q_1	1	2

For each row, the Borcherds invariant of the product is its weight: $\kappa_{\mathrm{BKM}}(F_{t,N}) = \mathrm{wt}(F_{t,N})$*. The last column is the corresponding twice-weight tuple* $(10, 4, 6, 2, 4, 1, 3, 2)$*, not the CHL constant table* $(10, 8, 6, 4, 2)$.

Proof. Cléry–Gritsenko classify the reflective genus-two modular forms with diagonal divisor and obtain exactly these eight holomorphic products. The candidate $(2, 4; \frac{1}{2})$ is not holomorphic, so it is not a ninth row. Their Borcherds product theorem gives the product weight as one half of the cusp-weighted sum of the row's zeroth Jacobi coefficients; only the single-cusp cases reduce to a single coefficient $c(0,0)/2$. The CHL ladder instead uses the five weak Jacobi inputs ϕ_N, $N \in \{1, 2, 3, 4, 6\}$, producing Borcherds products Φ_N with constants $(10, 8, 6, 4, 2)$ and weights $(5, 4, 3, 2, 1)$. The two lists share only the $(1,1)$ product $\Delta_5 = \Phi_1$. The scalar-square identities that occur in the CHL/dyon literature are row-specific:

$$\nabla_3^2 = \Phi_6^{(1,2)}, \qquad \nabla_2^2 = \Phi_4^{(1,3)}, \qquad \nabla_{3/2}^2 = \Phi_3, \qquad Q_1^2 = \Phi_2^{(2,4)}.$$

They use their own levels, covers, branches, and denominator data. They do not identify the five-entry CHL table with the Cléry–Gritsenko catalogue, and they do not turn the twice-weight tuple $(10, 4, 6, 2, 4, 1, 3, 2)$ into $(10, 8, 6, 4, 2)$. □

Proposition 8 ($K3 \times E$ spectrum). *The four labelled* $K3 \times E$ *values are*

$$\begin{aligned} \kappa_{\mathrm{cat}}(K3 \times E) &= 0, \\ \kappa_{\mathrm{ch}}^{\mathrm{Heis}}(K3 \times E) &= 3, \\ \kappa_{\mathrm{BKM}}(\Phi_1) &= 5, \\ \kappa_{\mathrm{fiber}}(K3) &= 24. \end{aligned}$$

Thus the labelled spectrum is $\{0, 3, 5, 24\}$. *The labels are part of the assertion.*

Proof. Künneth gives

$$\Xi(K3 \times E) = (1 - 0 + 1)(1 - 1) = 0$$

and

$$\kappa_{\mathrm{cat}}(K3 \times E) = \chi(\mathcal{O}_{K3})\chi(\mathcal{O}_E) = 2 \cdot 0 = 0.$$

The Heisenberg–Mukai specialisation is additive on the K3 and elliptic branches, giving

$$\kappa_{\mathrm{ch}}^{\mathrm{Heis}}(K3 \times E) = \kappa_{\mathrm{ch}}^{\mathrm{Heis}}(K3) + \kappa_{\mathrm{ch}}^{\mathrm{Heis}}(E) = 2 + 1 = 3.$$

The $N = 1$ Borcherds row gives $\kappa_{\mathrm{BKM}}(\Phi_1) = c_1(0)/2 = 10/2 = 5$. The fibre invariant is

$$\kappa_{\mathrm{fiber}}(K3) = \mathrm{rk}\, H^*(K3, \mathbb{Z}) = 24.$$

The number 2 is the fibre categorical value $\chi(\mathcal{O}_{K3})$, not the total-space value $\kappa_{\mathrm{cat}}(K3 \times E)$. The equality $3 + 2 = 5$ is a coincidence in this row, not a formula for κ_{BKM}. □

Proposition 9 (Igusa square normalisations). *Let Δ_5 denote the theta-constant normalisation of the Igusa square root, so that*

$$[q^{1/2}r^{1/2}s^{1/2}]\Delta_5 = 64.$$

Let D_5 be the monic Borcherds product attached to the $\phi_{0,1}$ input. Then

$$D_5 = 64^{-1}\Delta_5.$$

Oberdieck–Pixton's primitive scalar product squares this monic product:

$$\Phi_{10}^{\mathrm{OP}} = D_5^2 = 4096^{-1}\Delta_5^2.$$

In the quotient-by-E, Behrend-weighted $K3 \times E$ reduced DT/PT normalisation,

$$Z_{\mathrm{OP}}^X = Z_{\mathrm{DT}}^X = -\left(\Phi_{10}^{\mathrm{OP}}\right)^{-1} = -4096\,\Delta_5^{-2}.$$

The DVV/DMVV physics normalisation uses the unscaled Igusa square up to scalar; in the theta-normalised convention this is

$$\Phi_{10}^{\mathrm{un}} = \Delta_5^2.$$

Proof. The Borcherds–Gritsenko–Nikulin product supplies the monic product

$$D_5 = q^{1/2}r^{1/2}s^{1/2}\prod(1-q^n r^l s^m)^{f(nm,l)}$$

with vacuum coefficient 1, before expanding the real-root factor. The theta-constant normalisation of the same product has coefficient 64 at $q^{1/2}r^{1/2}s^{1/2}$. Hence $D_5 = 64^{-1}\Delta_5$. Squaring gives

$$D_5^2 = (64^{-1}\Delta_5)^2 = 4096^{-1}\Delta_5^2.$$

Oberdieck–Pixton's primitive scalar theorem gives the negative inverse of this monic square. The reduced DT equality uses the quotient-by-E, Behrend-weighted $K3 \times E$ comparison through reduced DT/PT and stable-pairs/Gromov–Witten correspondence. In that normalisation,

$$Z_{\mathrm{OP}}^X = Z_{\mathrm{DT}}^X = -(D_5^2)^{-1} = -4096\,\Delta_5^{-2}.$$

DVV/DMVV use the genus-two Igusa form up to scalar; the unscaled theta-normalised automorphic square is $\Phi_{10}^{\mathrm{un}} = \Delta_5^2$. The scalar $4096 = 64^2$ is a normalisation conversion, not a new Borcherds weight and not a Hall-orientation character. □

Remark 10. The compact Hodge trace, the Heisenberg–Mukai specialisation, the Borcherds weight, and the fibre rank are four constructions. The notation records this separation; erasing the subscripts erases the theorem.